

Examiner reports

Q1.

This question discriminated well between students of varying abilities. The majority of students understood that they had to integrate and chose the appropriate technique, often applying integration by parts successfully. A significant number of students made no attempt to go any further and completely ignored the constant of integration. These students generally achieved three marks.

Students who realised they had to determine the constant of integration often made good progress and scored 7 or 8 marks. Those who dropped marks at this point generally showed insufficient detail to fully justify their answer. Several very thorough, complete solutions were seen.

Q2.

(a) was very well answered with around two-thirds of students scoring full marks.

(b) proved to be quite a challenging question with a wide spread of marks. A number of students failed to see the connection to (a) despite the “Hence...” in the question, and so never arrived at an appropriate integrand to integrate. There were a wide variety of methods used to integrate including inspection and various substitutions which had the potential to be successful, and some very imaginative approaches were seen. Some students used a substitution of $x = 2\theta$ initially but failed to correctly replace $d\theta$, meaning that, despite correct later work, they were unable to arrive at a valid, correct solution.

Q5.

In part (a), apart from small number of students who did not differentiate the expression as

a product of two functions and obtained gradient functions such as $\frac{dy}{dx} = -8x \sin 2x$, hence gaining no marks, this question was usually answered very well. Some students failed to simplify their expression for the gradient after substitution of $\frac{\pi}{4}$, thus giving rise to some very complicated forms for the equation of the line.

In part (b), almost all the students demonstrated they could integrate by parts, with many gaining full marks. Where marks were lost it was usually due to an incorrect coefficient

written in the integration of $\frac{dy}{dx} = \cos 2x$ with $v = 2\sin 2x$ being common. Some students failed to simplify their final expression and there were some cases where a non-exact decimal value was written as an answer. A small number of attempts at the volume instead of the area were seen.

Q6.

In part (a), the integration proved difficult for many students who gave wrong coefficients

such as $\frac{1}{2}$, 2 or $\frac{1}{1-2}$ and incorrect expressions such as $-\frac{1}{1-2x} e^{1-2x}$. The use of

substitution often did not lead to a correct result, with $\frac{e^u}{u}$ being common. Many students

found the elimination of $\ln$ in $e^{-2\ln 2}$ a challenge. Some students mistakenly thought that changing their result into a decimal number and showing that it approximated $\frac{3}{8}e$ was a sufficient proof.

In part (b), most students gained the first method mark for giving $\frac{du}{dx} = \sec^2 x$ and the B mark for either changing the limits or using the original limits after their integration. The replacement of dx with $\frac{1}{\sec^2 x} du$ was sometimes seen for the first accuracy mark. The identity $\sec^2 x = 1 + \tan^2 x$ was often seen but replacing $\sec^2 x$ in terms of u often produced $(1 + u)$ rather than $(1 + u^2)$. Many students thought that they could integrate an expression that was a mixture of u and $\tan x$ terms. A surprising number of able students had difficulty simplifying $(1 + u^2)$

$$\sqrt{u} \text{ to } \left(u^{\frac{1}{2}} + u^{\frac{5}{2}} \right)$$

Q7.

A full spread of marks was seen on this question, with about 20% of the entry gaining full marks, and about another 20% only scoring 2 marks or fewer.

Part (a). Many students attempted this integral by using parts and produced some complicated looking expressions using incorrect integrals. This integral cannot be done by parts; some students did realise this and swapped to the required substitution method. Those who did use an appropriate substitution were usually successful, with an occasional error being made in the coefficient. Few students could do the integral by inspection.

Part (b). Most students knew they had to separate the variables and integrate both sides, with the vast majority using their result from part (a). Most used correct notation in the separation, but some students read e^{2y} as e^2y and some did not multiply across correctly by e^{2y} . Of those who did, most solved the integral of e^{2y} correctly. Those who also solved the integral in x correctly usually went on to complete successfully, although the common error was to think $\ln(a + b) = \ln a + \ln b$. Students who did use their result from part (a), whatever it was, could score method marks for using $(1,0)$ to find a value for their constant, and to use logarithms to express their final answer as $y = f(x)$.

Most students did include a constant, and did attempt to take logs to obtain a final solution.

Q8.

- (a) (i) The great majority of the students made progress here, most integrating correctly but some losing the final accuracy mark because of the omission of the constant of integration which was needed for the indefinite integral.
- (ii) There were three main approaches here and the most popular was to use $u = \ln x$, $\frac{dv}{dx} = \ln x$. If the answer to part (i) was correct often all the marks were

scored but sometimes there was a sign error in the final line. Using $u = (\ln x)^2$, $\frac{dv}{dx} = 1$ was also successful for many students, although the omission of brackets on the first term lost the accuracy mark. Some students were not able to differentiate $(\ln x)^2$ correctly. The third approach was using substitution for $\ln x$, but not many were able to follow that through and revert to a function of x . There were some poor attempts such as u

$$= (\ln x)^2, \quad \frac{du}{dx} = \frac{1}{x^2} \quad \text{and} \quad \int (\ln x)^2 dx = \frac{(\ln x)^3}{3}.$$

- (b) Most students correctly made the initial step of finding $\frac{du}{dx}$ correctly. A small number of students wisely set $x = u^2$ and found $\frac{du}{dx}$. Writing the integrand in terms of u was found to be very challenging by many students and numerous attempts at falsely cancelling $x^{\frac{1}{2}}$ prior to substitution often resulted in expressions that gained no further credit. Many who did get to the required expression failed to move on as they did not cancel by u and thought they could integrate to $\ln(u^2 + u)$. The more fundamental errors were $\frac{u}{u^2 + u}$ becoming $\frac{1}{u^2 + 1}$ or $\frac{1}{u^2} + 1$. Students should be made aware that the whole expression needs to stay within the integral; some had $2u$ or the equivalent outside the integral. Some students who got to the last line used the x limits in the u expression.

Q9.

There was a full spread of marks seen in this question with about a quarter of the students scoring 8 or more of the 10 marks available, but a similar proportion scoring 2 or less.

- (a) Most students recognised that this required integration by parts, although some just 'integrated' the t term and the cosine term independently, for no credit. Some tried to use substitution, and indeed some students substituted $u = \frac{\pi t}{4}$ and then used parts. Most identified the parts the right way round but those who did not should have found the integral becoming impossibly harder but they were often determined to get a result, again for no credit. Of those who started correctly, many made a mistake in writing down the parts formula, often confusing the sine and cosine terms. Many had the required coefficient $\frac{4}{\pi}$ inverted. Many who had the coefficients correct at this stage then made an error when integrating the sine term, often getting $\frac{8}{\pi}$ or similar. Many made a mistake when simplifying a correct expression, which was not penalised until part (b).
- (b) Most students separated the variables correctly, and usually used the correct notation. Occasionally the x was dropped or the dx or dt did not appear or was in the wrong place. A few attempted to 'integrate' the expression as given in the question to produce answers of no value. Most gained the mark for integrating x correctly, although some used logarithms due to incorrect separation. Most of the students also gained the method mark for using their result from part (a) in an attempt to find a constant of integration. Some students just substituted $t = 0$, $x = 4$ into the

differential equation itself, which made no sense. Those who had the correct form of the integral from part (a) could then score the remaining method mark for calculating firstly x^2 and then x . Many were correct to this point and then misunderstood the request to give their answer to the nearest centimetre or they evaluated using degrees rather than radians. About 8% of the entry gave a fully correct solution.

Q10.

In part (a), almost everyone earned the mark.

Part (b) was answered very well. Most candidates who realised the numerator was related to the derivative of the denominator and hence required a $\ln$ function were successful in obtaining full marks. There were two very common errors:

- candidates losing the final accuracy by writing $\frac{1}{6} \ln 91 - \frac{1}{6} \ln 21 = \frac{1}{6} \ln 70$
- candidates losing 2 marks by obtaining an incorrect value of k ; the most common value was 6 although $\frac{1}{2}$ and $\frac{1}{3}$ were also seen.

The major errors seen were expressions such as $\frac{2x^2 - 1}{12x^2 - 6} \ln(4x^3 - 6x + 1)$. Some candidates were able to recover by simplifying to the correct expression $\frac{1}{6} \ln(4x^3 - 6x + 1)$ before substitution of the limits; however some candidates substituted for 3 and 2 through the whole unsimplified expression and therefore gained no credit.

Q11.

In part (a), it was essential to use the quotient rule, as requested – it is specifically mentioned in the specification – and to show sufficient steps in the proof of the given answer. It was surprising that quite a number of candidates were unable to go from $\frac{-\cos \theta}{\sin^2 \theta}$ to the final expression.

In part (b), it was good to see some completely correct solutions though some used 30° and 45° , which was incorrect. A substantial number of candidates who correctly started the question lost a minus sign in their work.

Many candidates fell at the first hurdle by not replacing dx by an expression in θ and $d\theta$.

There were B marks available for a correct $\frac{dx}{d\theta}$, for $\sqrt{\operatorname{cosec}^2 \theta - 1}$ converted to $\sqrt{\cot^2 \theta}$ and for the limits in radians, but even these marks were often not earned. Unfortunately, many confused $\frac{dx}{d\theta}$ with $\frac{d\theta}{dx}$ and of course made no headway after that. Candidates who produced the correct numerical answer from their calculators without correct working did not gain marks.

Q12.

A substantial proportion of candidates solved this differential equation successfully, which is an impressive achievement given that there is a considerable amount of careful manipulation involved. Most candidates started with a correct separation, with most of

these using correct notation. Many integrated $\frac{1}{y^2}$ correctly, but this was not the case with the integration of $x \sin 3x$. Some showed no detail of an attempt at parts and scored no further marks. Of those who did show stages in their parts integration, some made sign and / or coefficient errors. Most candidates knew that they were to include a constant and attempted to find it, but they needed to have a $\cos 3x$ and a $\sin 3x$ term in their expression

to demonstrate knowledge of $\sin \frac{\pi}{2} = 1$ and $\cos \frac{\pi}{2} = 0$ when finding the value of the constant. The final stage of factoring out a 9 and inverting was done successfully by most who had got this far, although some incorrectly added the constant after inverting and scored no further marks. Those who failed to gain the last mark usually made an error in their last line or two of working, making a sign error or dropping a 3 or even the x .

Q13.

In this question, students tended to do very well or very badly.

It was good to see that many students were able to complete this integration correctly. However there were many blank scripts, and the majority of students obtained only the B

mark for a correct answer for $\frac{du}{dx}$. Some students were able to make slight progress, and where M1 was obtained, many did manage to produce some form of $\ln$ function, although

many made a sign error and obtained $\ln u - \frac{2}{u}$. There were also many fully correct solutions seen.

Q14.

For part (a), some student wrote down the correct differential equation apparently fully understanding all the information given and interpreting it correctly. However, all sorts of errors abounded in other attempts, some not even involving a derivative, and some with derivatives in x and y . Some attempts bore little or no relation to the information given, although credit was given if $\pm (2 - h)$ appeared in an otherwise incorrect expression. Many had a spurious t and / or h , either as a multiple or power, and the k appeared in a variety of places. Some students did not even form an equation, leaving a proportionality sign in their answer.

In part (b), most students knew they were expected to separate the variables and did it correctly, although there were some notation errors in the positioning of dx , at the front rather than the rear of the integrand. Those who failed to separate the variables, just produced nonsense.

Most students integrated $k dt$ successfully, but relatively few made good progress with the integral in x . No guidance was given, but students split fairly evenly between attempts by parts and by substitution, with similar success, or lack of it, via either method. Many students just integrated each term in the product; presumably this was an attempt at parts. Those who showed the parts and how the integral might be achieved scored a method

mark, but many could not integrate $\sqrt{(2x-1)}$ correctly, making an error in the required coefficient.

Those who used substitution usually chose to put $u = 2x - 1$ but did not always change their whole integral to be in terms of u ; a missing du was penalised for the method mark. Some with a correct first stage then did not multiply out to obtain an integrable form, but just integrated each term in the product. All students, no matter what their attempt at the integral, could obtain a method mark if they included a constant and tried to find it using the given initial conditions. Those few who did integrate and find the constant correctly, usually went on to calculate the final requested value of t correctly.

Q15.

Part (a) was very well answered, with the majority of candidates obtaining the correct function. The main error made by candidates who obtained a $\ln$ function was with the multiplying constant, which was often 2 or was completely missing. Omission of brackets was also a problem.

Not many of the candidates answered part (b) fully correctly. Only a few integrated by parts 'the

wrong way round' and some weaker candidates put $u = x$ and $v = \sin \frac{x}{2}$. However most knew

that they should integrate the $\sin \frac{x}{2}$ term, but did it wrongly, getting $\pm \cos \frac{x}{2}$ or more commonly

$\pm \frac{1}{2} \cos \frac{x}{2}$. If the first method mark was earned, the second was generally scored as well.

Q16.

Almost everyone earned the mark for differentiating $1 + 2\tan x$ correctly. It was good to see some excellent fully correct solutions, but many candidates made no further progress. Some had problems dealing with the 2, either at the substitution stage or when trying to simplify in order to integrate. Many failed to recognise that $\cos^2 x \times \sec^2 x$ equals 1. It was disconcerting to see these terms jumping from one side of the integral sign to the other on some scripts. Some tried to write these expressions in terms of u and did not progress further. A few failed to substitute back into an expression in x at the end.

Q17.

Part (a) was well answered by many of the candidates. The majority of candidates differentiated the x and integrated the sine function. Those candidates who made an error in the integration were able to gain the method marks. Those candidates who gained the first accuracy mark usually went on to give fully correct solutions. Where candidates gained no marks, it was usually due to setting up the parts incorrectly: it was common to see $u = x \sin$ and $v = 2x - 1$.

In part (b), most candidates made the appropriate start, with $\frac{du}{dx} = 2$. Although fully correct responses were seen, this question part was not very well answered by many candidates. It was essential to substitute for dx , $(2x - 1)$ and x^2 in the integral to earn the second method mark and some omitted the first of these or the integral sign. In putting x^2 in terms of u there were several hurdles: some only substituted for x , many forgot to square the 2 on the denominator, many had $(u - 1)$ instead of $(u + 1)$ and many only had two terms when they attempted to square. Work correct up to this point often did not go any further.

Q18.

Very few candidates achieved completely correct solutions to this question.

Part (a) was well answered by the majority of candidates. The main error was the two terms in the numerator being interchanged, resulting in $k = +12$. Other errors came from the expansion of the brackets, with various numerators including $8x$, and with -3 not being multiplied by 4, giving a final constant of -3 .

In part (b)(i), the expression $\ln(4x - 3)$ was rarely differentiated completely correctly. A

common error was to get $\frac{x}{4x-3} + \ln(4x - 3)$, rather than $\frac{4x}{4x-3} + \ln(4x - 3)$, which earned two method marks.

In part (b)(ii), $\frac{dy}{dx}$ was frequently calculated correctly, following through from part (b)(i), but then this value was used as the gradient of the normal, with some candidates explicitly stating that they were finding the normal.

In part (c)(i), most of the candidates earned a method mark for ' $du = 4dx$ '. Many of these candidates went on to write the integral completely in terms of u . Some candidates lost

marks at this stage through either obtaining an extra factor of 4 or getting $x = \frac{u-3}{4}$ when rearranging $u = 4x - 3$.

Although many fully correct responses were seen, some candidates were unable to proceed beyond this point: part (c)(ii) proved to be beyond many candidates, as the correct application of integration by parts was not common. Most candidates started the

question with $u = \ln$ and $\frac{dv}{dx} = 4x - 3$. Those candidates who correctly started this part of the question often failed to get the first accuracy mark because of their derivative of $\ln(4x - 3)$, losing the factor of 4 in the numerator. However, many were able to get the next method mark because they correctly used their value from part (c)(i). A few candidates spotted that the result obtained in part (b)(i) could provide an alternative method of solution.

Q19.

Most candidates lost marks on this question. Full marks were not common although they

were seen from the more able candidates.

Not many candidates obtained full marks for part (a). Most candidates scored 2 out of 4 since they started correctly by differentiating $\ln t$ and integrating $(t - 1)$. Obvious errors were differentiating both terms or starting by trying to integrate $\ln t$. Several candidates managed to obtain the second accuracy (A) mark for simplification to $(t^2/2 - t) \ln t - \int (t/2 - 1)dt$ but lost the final accuracy mark by ending up with a final term of $-t$, not $+t$.

Part (b) was done well by most candidates, although poor manipulation often cost the loss of the final accuracy mark. Some candidates also confused the issue by trying to introduce terms in u and du .

Part (c) was not very well answered, with candidates failing to change the limits as a common error. Many candidates did not appreciate that they should be using their answer to part (a) and tried to start again, often obtaining different answers to those they had found in part (a). After correctly approaching the choice of u and dv/dt , most were then defeated by the required manipulation of the subsequent algebra.

Q20.

Part (a)(i) was well answered.

Candidates who saw the connection with part (a)(i), usually made a very good attempt at part (a)(ii), often with complete success. Limits did cause a problem where candidates had not bracketed $2x^2 - 8x + 3$, and some candidates did not leave the answer in the required

form but left their answer as $\frac{1}{4} \ln 9$.

Part (b) was well answered by candidates who were able to write the integral in terms of u successfully. Many candidates lost a mark through omission of either du or c . A disturbing

number of candidates thought that $\frac{1}{3} \times \frac{1}{3} = \frac{1}{6}$. Some clearly able candidates lost the last mark by spoiling an otherwise acceptable answer.

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Q21.

Part (a) of this question was reasonably answered with most candidates obtaining the method mark for the product rule. Some candidates then lost the accuracy mark through incorrect evaluation of the constant associated with the derivative of $\cos 2x$; 2 and $\frac{1}{2}$ were frequently seen.

Answers to part (b)(i), although frequently correct, were often badly set out with the function being equated to zero or x being changed to α at various points in the solution.

Division of $\cos 2x$ by $\cos 2x$ often resulted in zero before going on to fudge the correct answer. Part (b)(ii) was usually well answered with correct evaluations of $f(0.4)$ and $f(0.5)$. Some candidates then lost marks by just saying α was 'between these two values' without stipulating 0.4 and 0.5. Part (b)(iii) was usually well done but there were many cases when

the $\tan$ became removed from its $2x$ and $\frac{1}{2} x \tan^{-1} = x$ was often seen. Part (b)(iv) was generally well answered but marks were lost from the use of degrees and for not writing

the answer to the required degree of accuracy.

Many fully correct answers and many answers which only lost the final accuracy mark were seen in part (c). Other answers which had the wrong coefficient associated with the $\sin 2x$ often got the method marks. There were candidates who started with $u = x$ and $v =$

$\cos 2x$ or started with $\frac{dv}{dx} = x$ and obtained a more difficult integral.

Q22.

Part (a) was very well answered with most candidates gaining all 3 marks but it was not uncommon to see an incorrect quotient formula used (usually the wrong order in the numerator, but sometimes the wrong denominator)

Part (b) was reasonably well answered. However a number of candidates tried to tackle this question by squaring, generally without success.

There were very few successful attempts seen in part (c). Most candidates failed to

provide $\frac{dx}{d\theta}$ at all. Many of those who did make an attempt chose to try a different substitution of their own, either immediately or subsequently.

Q23.

Many candidates answered part (a) correctly and scored full marks but there were a significant number who gave a denominator of 10 in the second term of the answer after an otherwise correct approach. Some candidates tried to integrate x and differentiate e^{5x} which thereafter gave them no further credit.

In part (b)(i), most candidates obtained marks for finding $\frac{du}{dx}$ but their attempts to obtain the given answer were often flawed. Part (b)(ii) was not done particularly well; many split the denominator, and did not see that it required the use of $\ln$ at all.

Q24.

- (a) (i) This was reasonably well done, although some candidates integrated.
- (ii) Many candidates failed to appreciate the significance of part (a)(i) and obtained complicated incorrect expressions. Those candidates who did appreciate the relevance of part (i) either obtained the correct answer or $2\ln |x^4 + 2x|$.
- (b) (i) This was fairly well answered. However, many candidates lost marks because they did not realise that in any substitution question, apart from replacing functions of x with functions of u , they have to find and use $\frac{du}{dx}$.
- (ii) The integration was very well answered by the majority of candidates with

many accurately completing the question. Problems did arise through the use of incorrect limits. Many candidates converted their integration answers in u to terms in x and then used the original limits. This method was equally successful.

Q25.

- (a) Reasonable attempts were made by most candidates. The main mistake was finding the derivative of x^{-2} which was often seen as $-2x^{-1}$. This lost marks since candidates then 'found' the answer given by incorrect means.
- (b) There were many fully correct answers but many candidates chose incorrect functions for u and $\frac{dv}{dx}$, with the obvious implications.
- (c)
 - (i) This was reasonably well answered, the main error seen being $1 - 21nx = x^3$.
 - (ii) The majority of candidates obtained the method mark following through from their answer to part (b). Few candidates completed this question correctly.

Q26.

Part (a) This was very well answered by the majority of candidates. Where errors occurred it was where candidates left their answer in terms of u or failed to multiply by the derivative of $3x$.

Part (b) This question proved difficult for many of the candidates although fully correct responses were seen. Most were able to achieve part marks for $du/dx = 2$ and many realised they had to write the integral completely in terms of u . It was at this stage factors of $1/2$ were missing or dx was incorrectly replaced by $2 du$. A few candidates who correctly evaluated the integral in terms of u failed to put the expression back in terms of x .

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Q27.

Part (a) was well answered, although some candidates integrated. There were many fully correct answers to part (b), but a lot of candidates chose the incorrect functions to equal u and dv/dx with the obvious ensuing problem.

Part (c) was poorly answered. Candidates must realise that, in any substitution question, apart from replacing functions of x with functions of u , they must also find du/dx . A large number of candidates attempted to integrate an expression containing both x 's and u 's.